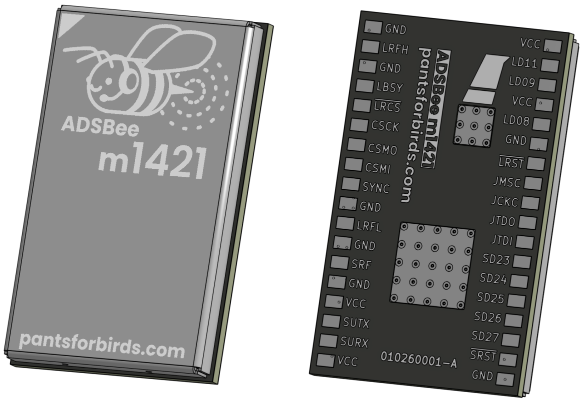
ADSBee m1421

Ultra Low-Power Dual Band ADS-B Receiver Module



Features

- 1090MHz Mode S and ADS-B packet decoding.
- Tunable sub-GHz transceiver for UAT / additional protocols.
- Multiple output formats over UART: Raw Frames / ADSBee CSV / MAVLINK1 / MAVLINK2 / Mode S Beast / GDL90 / Aircraft JSON.
- Firmware updates over UART or JTAG.

Applications

- Aircraft detect and avoid for UAS
- ADS-B In for low-power systems

Quick Specs

Power Draw	19mA* @ 1.8-3.3V
Minimum RF Input Power Level	Mode S: -90dBm* UAT: -87dBm*
Dynamic Range	Mode S: 90dBm* UAT: 87dBm*
Simultaneous Aircraft Tracks Supported	≤400
Weight	1.85g

* = subject to change during validation

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Revision History

20260606

- Documented the serial command interface and reporting protocols. Renamed the 1090 MHz commands to the R1090_ prefix and added RX_STATS.

20260516

- Initial release.

20260606

- Add detailed protocol descriptions.
- Revise power estimate.

1 Overview

ADSBee m1421 is the world's lowest power and most cost-effective dual-band ADS-B receiver. The name comes from what's under the hood: a TI CC13**14** low power RF MCU and a Semtech LR20**21** radio. These two ICs combine the best of two worlds: the LR2021's best in class sensitivity for high baud rate OOK RF protocols, plus the CC1314's low-power compute and high sensitivity sub-GHz RF reception.

ADSBee m1421 is built with open-source firmware from Pants for Birds LLC's ADSBee project, and includes onboard demodulation and decoding for Mode S and UAT messages, with ADS-R, TIS-B, and FIS-B (including NOTAMs and weather). An onboard aircraft dictionary contains all relevant information for up to 400 aircraft targets, and reports target information via UART using a wide range of available protocols, including RAW, CSBee, MAVLINK1, MAVLINK2, GDL90, and Mode S Beast.

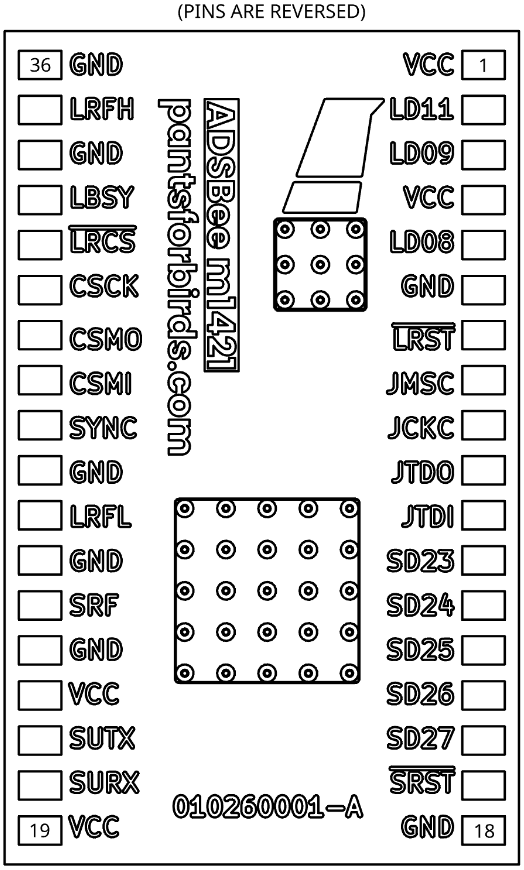
Customers who want direct control of the onboard LR2021's 2.4GHz radio (e.g. for LoRa or mesh radio applications) can suspend the CC1314 using the SYNC line and configure the LR2021 directly over SPI. All LR2021 SPI pins, plus three GPIOs, are provided for direct interfacing with the radio (interrupts, etc).

2 Pin Functions

Pin #	Pin Name	Pin Function	Pin #	Pin Name	Pin Function
1	VCC	Power supply (1.8V – 3.3V).	19	VCC	Power supply (1.8V – 3.3V).
2	LD11*	LR2021 DIO 11.	20	SURX	SubGHz MCU UART RX.
3	LD09*	LR2021 DIO 12.	21	SUTX	SubGHz MCU UART TX.
4	VCC	Power supply (1.8V – 3.3V).	22	VCC	Power supply (1.8V – 3.3V).
5	LD08*	LR2021 DIO 8.	23	GND	Ground.
6	GND	Ground.	24	SRF	SubGHz RF connection.
7	~LRST*	LR2021 Reset (active low).	25	GND	Ground.
8	JMSC	SubGHz MCU JTAG TMSC.	26	LRFL	Mode S RF connection.
9	JCKC	SubGHz MCU JTAG TCKC.	27	GND	Ground.
10	JTDO	SubGHz MCU JTAG TDO.	28	SYNC*	SubGHz sync (drive high to suspend SubGHz MCU).
11	JTDI	SubGHz MCU JTAG TDI.	29	CSMI*	LR2021 SPI MISO.
12	SD23	SubGHz MCU DIO 23.	30	CSMO*	LR2021 SPI MOSI.
13	SD24	SubGHz MCU DIO 24.	31	CSCK*	LR2021 SPI SCK.
14	SD25	SubGHz MCU DIO 25.	32	~LRCS*	LR2021 SPI CS.
15	SD26	SubGHz MCU DIO 26.	33	LBSY*	LR2021 busy.
16	SD27	SubGHz MCU DIO 27.	34	GND	Ground.
17	~SRST	SubGHz MCU Reset (active low).	35	LRFH*	LR2021 2.4GHz RF connection.
18	GND	Ground.	36	GND	Ground.
MP		Thermal pad for CC1314 and LR2021.			

*= Only required for direct control of LR2021. Leave unconnected for normal applications.

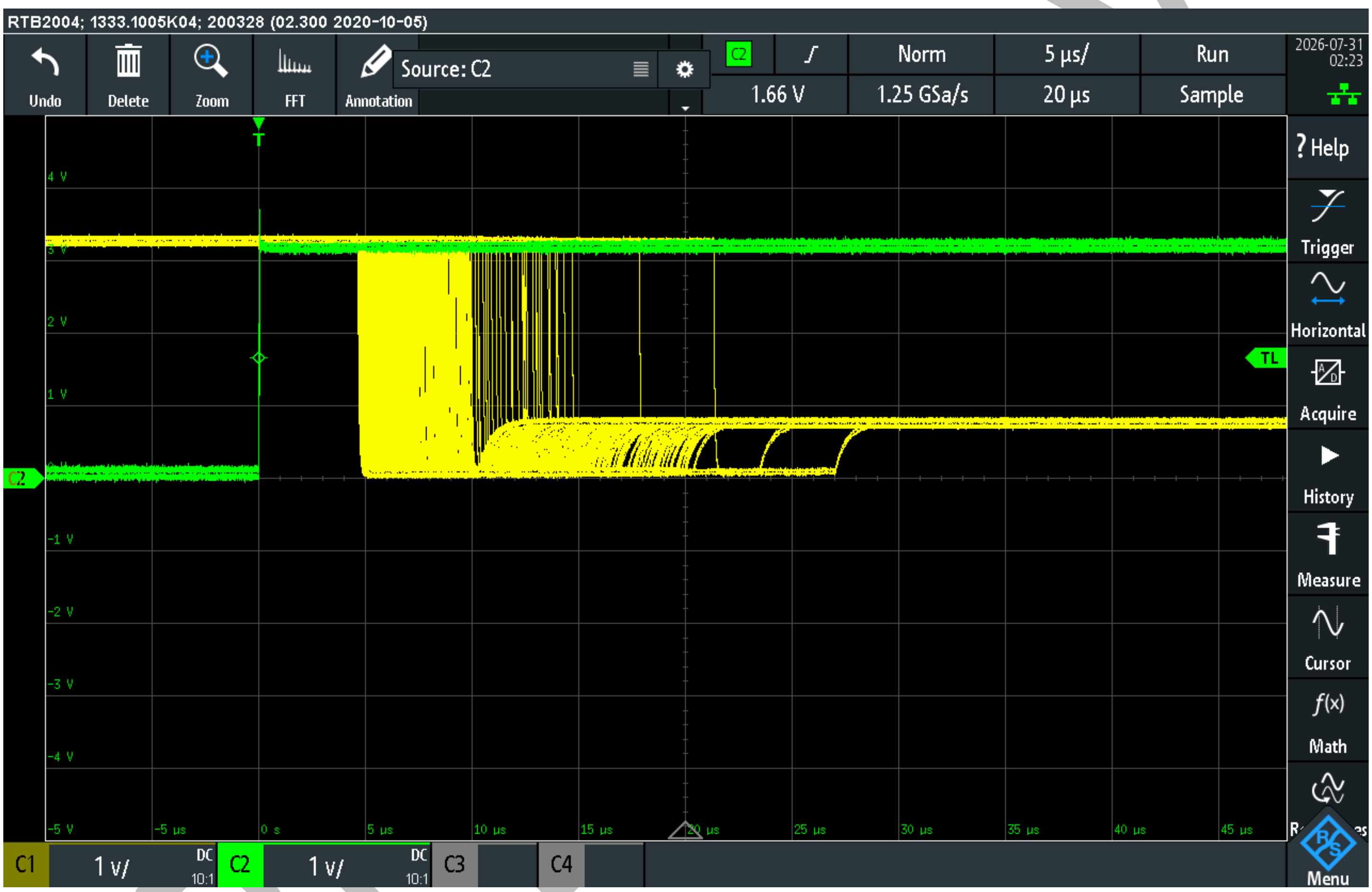
MODULE BACK SIDE
(PINS ARE REVERSED)



2.1 SYNC Pin Functionality

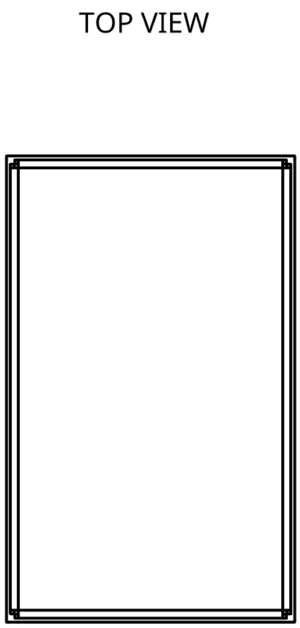
The SYNC pin can be driven high to put the CC1314R10 into a low-power sleep state, with all LR2021 control lines tri-stated in order to allow external control of the LR2021.

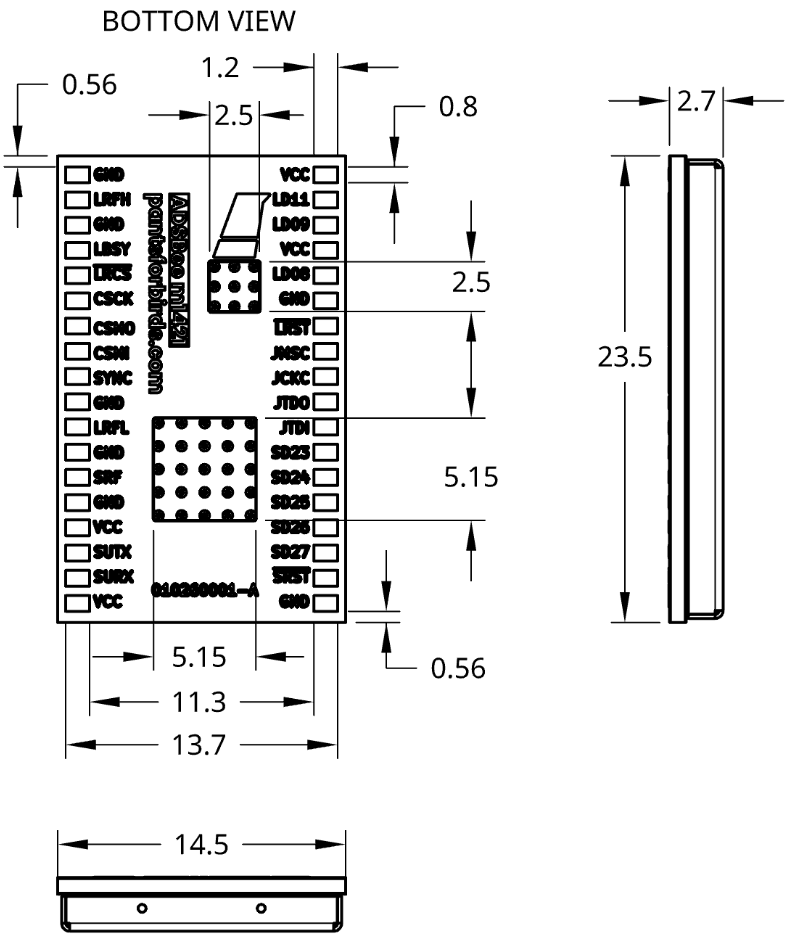
The green trace below shows the SYNC line going high, and the yellow trace shows the LRST (LR2021 RESET) net going to tri-state, in a repeated test with trace persistence turned to maximum. To be safe, an external MCU may assume that the LR2021is freed 40us after asserting the SYNC line high, or after the LRST line is tri-stated, whichever occurs first.



3 Device Form Factor

3.1 Module Dimensions





NOTE:
Dimensions in mm.

All vias shown on module
are epoxy filled and capped.

3.2 Recommended Footprint

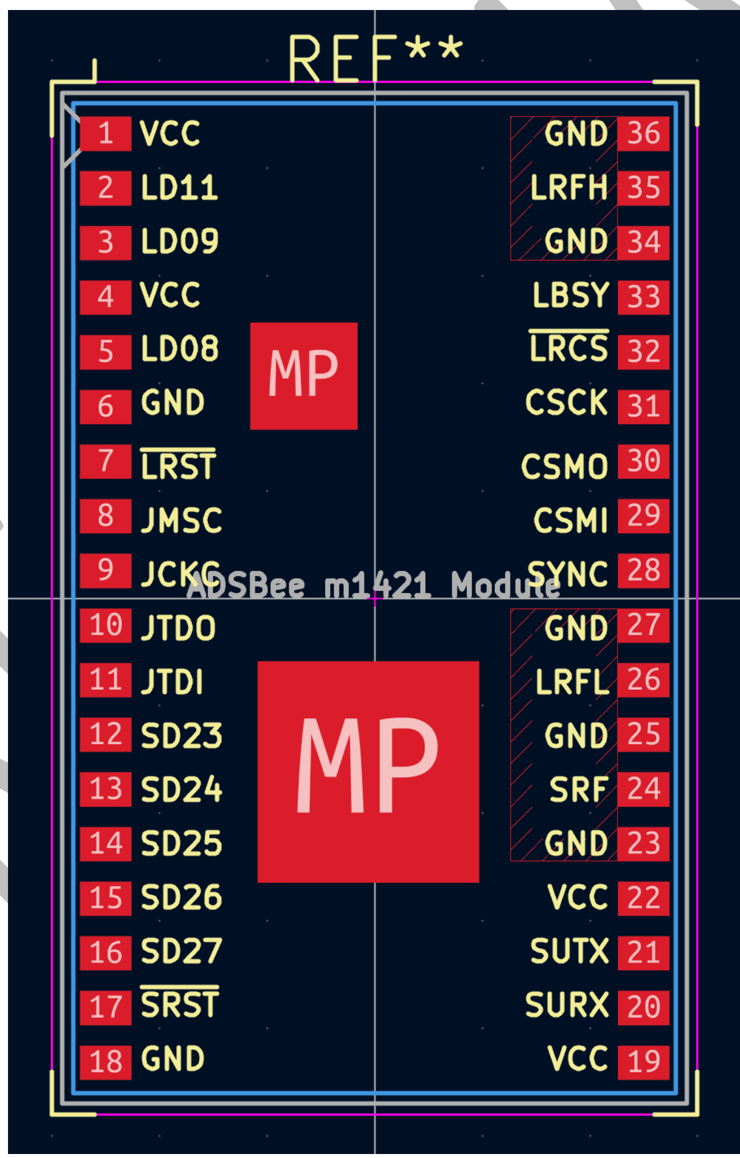
A KiCAD footprint is available at: https://github.com/CoolNamesAllTaken/kicad-libs/blob/main/lib_fp/Custom_Module.pretty/PantsForBirds_ADSBee_m1421_Module.kicad_mod

A screenshot of the footprint is included below. Note the top copper keepouts (hatched red areas) where impedance controlled transmission lines exist on the surface of the module.

LRFH, LRFL, and SRF are coplanar waveguides controlled to 50 Ohms. They can be broken out as microstrip or coplanar waveguide, depending on designer preference. Inserting filled and capped vias directly under the GND pads on the right side of the module is recommended for minimal impedance discontinuity.

LR2021 and CC1314 IC mounting pads are brought through the module using thermal vias. It is recommended to connect both pads labelled "MP" to a ground plane using filled and capped thermal vias.

All exposed copper pads shown in red should have paste.



4 AT Commands

ADSBee m1421 is configured over its Console UART using ASCII AT commands. Each command begins with the prefix AT+ and ends with a carriage return. Most commands provide a query form (AT+COMMAND?) that reports the current value and a set form (AT+COMMAND=<args>) that changes it. Changes take effect immediately; write them to nonvolatile memory with AT+SETTINGS=SAVE so they persist across reboots.

4.1 AT+BAUD_RATE

Changes the Console UART baud rate at runtime. The device acknowledges with OK at the old baud rate and then switches, so the host should reopen its serial port at the new rate after receiving the OK. The baud rate is not saved to nonvolatile settings: any reboot or power cycle returns the console to 115200 baud.

A baud rate of 921600 or above is recommended for best performance. Significant bottlenecks can occur if using verbose reporting protocols at the default baud rate of 115200.

4.1.1 Set the Console Baud Rate

```
AT+BAUD_RATE=CONSOLE,<baud>
OK
```

Parameter	Description	Acceptable Values
iface	Serial interface to configure.	CONSOLE
baud	New console baud rate.	115200, 230400, 460800, 921600, 1000000

4.1.2 Query the Console Baud Rate

```
AT+BAUD_RATE?
BAUD_RATE=CONSOLE,115200
```

4.2 AT+BOOT_UART_BOOTLOADER

Erases the application image and enters the CC1314 ROM UART bootloader (on pins DIO2/DIO3) for serial reflashing. The device remains in the bootloader until it is reflashed. A confirmation word is required to prevent accidental use.

```
AT+BOOT_UART_BOOTLOADER=1DEADBEE
```

Parameter	Description	Acceptable Values
confirmation	Confirmation word required to erase the app image and enter the bootloader.	1DEADBEE

4.3 AT+DEVICE_INFO

Reports manufacturing and firmware information about the module.

```
AT+DEVICE_INFO?
Part Code: ADSBEE-M1421
CC1314R10 Unique ID: 00112233AABBCCDD
CC1314R10 Firmware Version: 0.1.0
OTA Key 0:
OTA Key 1:
```

4.4 AT+LOG_LEVEL

Sets how much diagnostic output the firmware prints to the Console UART.

4.4.1 Set the Log Level

```
AT+LOG_LEVEL=<log_level>
OK
```

Parameter	Description	Acceptable Values
log_level	Console logging verbosity.	SILENT, ERRORS, WARNINGS, INFO

4.4.2 Query the Log Level

```
AT+LOG_LEVEL?
LOG_LEVEL=INFO
```

4.5 AT+MAVLINK_ID

MAVLink components have a system ID and a component ID. For more information about system and component IDs, visit https://mavlink.io/en/services/mavlink_id_assignment.html.

4.5.1 Set the MAVLINK System ID and Component ID

```
AT+MAVLINK_ID=<system_id>,<component_id>
OK
```

Parameter	Description	Acceptable Values
system_id	Unique identifier for the MAVLink system (e.g. the aircraft). All components in the system should share the same system_id.	1-255
component_id	Unique identifier for a component within the MAVLink system (e.g. the ADSBee). Each component should have a unique component_id.	1-255

Default: 156 (MAV_COMP_ID_ADSB).

4.5.2 Query the MAVLINK System ID and Component ID

```
AT+MAVLINK_ID?
System ID: 1
Component ID: 156
```

4.6 AT+PROTOCOL_OUT

Selects the reporting protocol used to output decoded data on a serial interface. See the Reporting Protocols section for the format of each protocol.

4.6.1 Set the Reporting Protocol for a Serial Interface

```
AT+PROTOCOL_OUT=<iface>,<protocol>
OK
```

Parameter	Description	Acceptable Values
iface	Serial interface to configure.	CONSOLE
protocol	Reporting protocol to use on the interface.	NONE, RAW, BEAST, BEAST_NO_UAT, BEAST_NO_UAT_UPLINK, CSBEE, MAVLINK1, MAVLINK2, GDL90, GDL90_NO_UAT_UPLINK, AIRCRAFT_JSON

4.6.2 Query the Reporting Protocol for All Serial Interfaces

```
AT+PROTOCOL_OUT?
PROTOCOL_OUT=CONSOLE,MAVLINK2
```

4.7 AT+R1090_GAIN

Sets the gain of the 1090 MHz front-end. Lower gain steps reduce sensitivity but improve dynamic range in high-signal environments. AUTO enables automatic gain control, which improves dynamic range at the cost of mangling the preamble. For best dynamic range, utilize AUTO gain with DF17 preamble. Defaults to AUTO.

4.7.1 Set the 1090 MHz Gain

```
AT+R1090_GAIN=<gain_step>
OK
```

Parameter	Description	Acceptable Values
gain_step	1090 MHz front-end AGC gain step.	0 or AUTO = automatic; 1-15 = manual (13 = maximum gain)

4.7.2 Query the 1090 MHz Gain

```
AT+R1090_GAIN?
R1090_GAIN=10
```

4.8 AT+R1090_PREAMBLE

Sets the preamble detection mode used by the 1090 MHz Mode S receiver. The DF17 mode triggers on the second preamble half plus the DF17 header bits, which provides best dynamic range. This comes at the cost of missing non-DF17 messages, including DF 4/5/11 (no squawk codes, pressure altitude from ADSB only, no all-call reply). For aircraft detection and collision avoidance purposes, DF17 preamble mode is recommended. Defaults to DF17. The MODE_S_SW_CRC mode uses the standard Mode S preamble with the hardware CRC filter disabled; the firmware validates the CRC in software, keeping failed-CRC frames available for error correction.

4.8.1 Set the Preamble Mode

```
AT+R1090_PREAMBLE=<mode>
OK
```

Parameter	Description	Acceptable Values
mode	1090 MHz Mode S receiver preamble detection mode.	MODE_S_PREAMBLE, DF17, MODE_S_SW_CRC

4.8.2 Query the Preamble Mode

```
AT+R1090_PREAMBLE?
R1090_PREAMBLE=MODE_S_PREAMBLE
```

4.9 AT+R1090_RX_BOOST

Sets the receive boost level of the LR2021 1090 MHz receive path. Higher boost levels increase receiver sensitivity at the cost of higher current consumption. Defaults to 0 (boost off).

4.9.1 Set the 1090 MHz RX Boost Level

```
AT+R1090_RX_BOOST=<boost>
OK
```

Parameter	Description	Acceptable Values
boost	1090 MHz RX path boost level.	0 = off; 1-7 (7 = maximum boost)

4.9.2 Query the 1090 MHz RX Boost Level

```
AT+R1090_RX_BOOST?
R1090_RX_BOOST=0
```

4.10 AT+REBOOT

Reboots the device.

```
AT+REBOOT
```

4.11 AT+RX_ENABLE

Enables or disables the onboard receivers. The first argument, if present, overrides the others, allowing all receivers to be enabled or disabled at once.

4.11.1 Enable or Disable the Receiver(s)

```
AT+RX_ENABLE=<all>,<1090>,<subg>
OK
```

Parameter	Description	Acceptable Values
all	Enables or disables all receivers. Overrides the other arguments if present.	0, 1
1090	Enables or disables the 1090 MHz receiver.	0, 1
subg	Enables or disables the Sub-GHz receiver.	0, 1

4.11.2 Query Whether the Receiver(s) Are Enabled

```
AT+RX_ENABLE?
1090 Receiver: ENABLED
SubG Receiver: ENABLED
```

4.12 AT+RX_POSITION

Sets or queries the receiver position and the source used to determine it. A receiver position is required for some calculations, such as the distance to a tracked aircraft.

4.12.1 Query Receiver Position

```
AT+RX_POSITION?
Receiver Position:
    Source: FIXED [OK]
    Latitude: 37.000000 deg
    Longitude: -122.000000 deg
    GNSS Altitude: 0 ft
    Barometric Altitude: 0 ft
    Heading: 0.0 deg
    Speed: 0 kts
    ICAO: 0x000000 [NOT IN USE]
```

4.12.2 Set Receiver Position

```
AT+RX_POSITION=<source>,<args>
```

Parameter	Description	Acceptable Values
source	Source used to determine the receiver position. Each source takes different additional arguments, described below.	NONE, FIXED, GNSS, LOWEST, ICAO

4.12.2.1 Disable Receiver Position

```
AT+RX_POSITION=NONE
OK
```

4.12.2.2 Set a Fixed Receiver Position

```
AT+RX_POSITION=FIXED,<lat_deg>,<lon_deg>,<gnss_alt_ft>,<baro_alt_ft>,<heading_deg>,<speed_kts>
OK
```

Parameter	Description	Acceptable Values
lat_deg	Latitude in degrees.	-90.0 to 90.0
lon_deg	Longitude in degrees.	-180.0 to 180.0
gnss_alt_ft	GNSS altitude in feet.	integer
baro_alt_ft	Barometric altitude in feet.	integer
heading_deg	Heading in degrees (true north).	0.0 to 360.0
speed_kts	Ground speed in knots.	>= 0

4.12.2.3 Use the GNSS Receiver Position

```
AT+RX_POSITION=GNSS
OK
```

4.12.2.4 Track the Lowest Aircraft

```
AT+RX_POSITION=LOWEST
OK
```

Bootstraps the receiver position from the lowest-altitude aircraft currently being tracked.

4.12.2.5 Track an Aircraft by ICAO Address

```
AT+RX_POSITION=ICAO,<icao>
OK
```

Parameter	Description	Acceptable Values
icao	ICAO address (hex) of the aircraft to use for position bootstrap.	0x000000 to 0xFFFFFF

4.13 AT+RX_STATS

Reports OOK receiver statistics from the LR2021 radio: received packet count (pkt_rx), CRC and length error counts (crc_error, len_error), preamble detector fires (pbl_det), sync word successes and failures (sync_ok, sync_fail), and receive timeouts (timeout). Also reports receiver health counters: LR2021 RX FIFO overflows (fifo_full), raw packet queue overflows (q_ovf), single-bit errors corrected by the decoder (bitflips_fixed), and UAT receiver error restarts (uat_rx_restarts). The counters accumulate until reset.

```
AT+RX_STATS?
RX_STATS=pkt_rx=1234,crc_error=5,len_error=0,pbl_det=1500,sync_ok=1234,sync_fail=12,timeo
ut=0,fifo_full=0,q_ovf=0,bitflips_fixed=42,uat_rx_restarts=0
```

Reset all statistics counters, including the health counters:

```
AT+RX_STATS=RESET
OK
```

4.14 AT+SETTINGS

Loads, saves, resets, or displays the device nonvolatile settings. Settings changed by other AT commands take effect immediately but are not retained across reboots until saved.

4.14.1 Query Current Settings

```
AT+SETTINGS?
```

Prints the current settings in a human-readable format.

4.14.2 Dump Settings in AT Format

```
AT+SETTINGS?DUMP
+PROTOCOL_OUT=CONSOLE,MAVLINK2
+R1090_GAIN=10
+RX_ENABLE=1,1,1
+WATCHDOG=0
...
```

Prints the current settings as AT commands that can be replayed to restore the configuration.

4.14.3 Load Settings from Nonvolatile Memory

```
AT+SETTINGS=LOAD
OK
```

4.14.4 Save Settings to Nonvolatile Memory

```
AT+SETTINGS=SAVE
OK
```

4.14.5 Reset Settings to Factory Defaults

```
AT+SETTINGS=RESET
OK
```

4.15 AT+WATCHDOG

The watchdog timer automatically resets the device if firmware execution stalls and the timer is no longer serviced, providing automatic recovery from a hard fault. It can be disabled (for example, during breakpoint debugging) and can be self-tested via AT command to verify that it is still active.

4.15.1 Configure the Watchdog Timer

```
AT+WATCHDOG=<timeout_sec>
OK
```

Parameter	Description	Acceptable Values
timeout_sec	Watchdog timeout in seconds. 0 disables the watchdog; 65535 is approximately 18.2 hours.	0-65535

4.15.2 Query the Watchdog Timer

```
AT+WATCHDOG?
WATCHDOG=10
```

Prints the watchdog timeout in seconds, or 0 if the watchdog is disabled.

4.15.3 Test the Watchdog Timer

```
AT+WATCHDOG=TEST
Blocking for 11 seconds.
```

Blocks the processor for the configured timeout plus one second, which should induce a reboot. If no reboot occurs, an error indicates that the watchdog failed to activate.

5 Reporting Protocols

ADSBee m1421 supports the following reporting protocols on CONSOLE.

- CSBee
- GDL90
- MAVLINK 1
- MAVLINK 2
- Mode S Beast
- Raw Packets
- Aircraft JSON

The CONSOLE interface reports debug messages and AT command responses in addition to the selected reporting protocol. If the CONSOLE interface is being used as a reporting interface, it is recommended to send AT+LOG_LEVEL=SILENT to silence any debug logs that might corrupt the reported data, and to avoid sending additional AT commands while reading reported data in order to avoid the reported data being interspersed with OK and other AT command responses from the ADSBee m1421.

5.1 CSBee

Comma Separated Bee protocol containing information about tracked aircraft as plain text.

The CSBee protocol is heavily inspired by the Aerobits Aero CSV protocol.

5.1.1 CSBee Protocol Version History

Consult the table below for a version history of the CSBee protocol. Each version of the ADSBee firmware writes only the most recent version of the CSBee protocol. All versions of the CSBee protocol except for the most recent version are considered deprecated. Whenever possible, efforts are made to reduce breaking changes to the CSBee protocol when incrementing the version number.

Current protocol version: 3

CSBee Protocol Version	Introductory Firmware Revision	Description
1		Initial release.
2	adsbee_1090-0.9.0-rc1	• Add UAT contacts. • Change bit flag positions. • Add dedicated fields for GNSS / baro vertical rates. • Add additional aircraft info (e.g. dimensions).
3	adsbee_1090-0.9.0-rc15	• Add is_non_transponder flag to Mode S aircraft. • Prepend address qualifier to UAT ICAO addresses. • Add is_utc_coupled flag to UAT aircraft.

5.1.2 Mode S Aircraft Message

This message contains information about an aircraft being tracked via ADS-B (1090MHz). Aircraft reports are provided once per second, per aircraft, until contact with the aircraft has been lost for 60 seconds.

```
#A:ICAO,FLAGS,CALL,SQUAWK,ECAT,LAT,LON,BARO_ALT,GNSS_ALT,DIR,SPEED,BARO_VRATE,GNSS_VRATE,
NICNAC,ACDIMS,VERSION,SIGS,SIGQ,SFPS,ESFPS,CRC\r\n
```

Field	Description	Format	Example value
#A	Mode S aircraft message start indicator.		
ICAO	ICAO number of aircraft (3 bytes).	Hex Integer	3C65AC
FLAGS	Flags bitfield, see table 5.1.2.1.	Hex Integer	12F356A8
CALL	Callsign of aircraft.	String	N61ZP
SQUAWK	SQUAWK of aircraft.	Octal Integer	7232
ECAT	Emitter category, see table 5.1.2.2.	Integer	14
LAT	Latitude, in degrees.	Float	57.57634
LON	Longitude, in degrees.	Float	17.59554
BARO_ALT	Barometric altitude, in feet.	Integer	5000
GNSS_ALT	Geometric altitude, in feet.	Integer	5000
DIR	Direction of aircraft, in degrees [0,360). Consult bit flags to determine whether this field is true heading, magnetic heading, or track.	Integer	35
SPEED	Horizontal velocity of aircraft, in knots.	Integer	464
BARO_VRATE	Barometric vertical velocity of aircraft, in ft/min.	Integer	-1344
GNSS_VRATE	Geometric vertical velocity of the aircraft, in ft/min	Integer	-1344
NICNAC	Aircraft data integrity. See table 5.1.2.4	Hex Integer	
ACDIMS	Aircraft physical dimensions and system design assurance. See table 5.1.2.5.	Hex Integer	31BE89F2
VERSION	ADS-B protocol version used by the aircraft.	Integer	2
SIGS	Signal strength, in dBm.	Integer	-92
SIGQ	Signal quality, in dB.	Integer	2
SFPS	Number of valid 56-bit Squitter Mode S frames received from the aircraft during the last second.	Integer	2
ESFPS	Number of valid 112-bit Extended Squitter Mode S frames received from the aircraft during the last second.	Integer	5
CRC	CRC16 (described in 5.1.2.6).	Hex Integer	2D3E

5.1.2.1 FLAGS Bitfield

Note: All bits 25-31 are momentary (cleared and updated every reporting interval).

Bit	Bit Name	Meaning if the bit is set (1)
0	IS_AIRBORNE	Emitter is airborne.
1	BARO_ALTITUDE_VALID	Emitter has provided a barometer altitude.
2	GNSS_ALTITUDE_VALID	Emitter has provided a GNSS altitude.
3	POSITION_VALID	Emitter has provided a pair of even and odd Compact Position Reporting packets that were decoded to a location.
4	DIRECTION_VALID	Emitter has provided a direction (may be ground track, magnetic compass heading, or true compass heading).
5	HORIZONTAL_SPEED_VALID	Emitter has provided a horizontal velocity.
6	BARO_VERTICAL_RATE_VALID	Emitter has provided a barometric vertical velocity.
7	GNSS_VERTICAL_RATE_VALID	Emitter has provided a GNSS vertical velocity.
8	IS_MILITARY	Emitter has transmitted at least one packet using a military format, such as Military Extended Squitter (DF=19).
9	IS_CLASS_B2_GROUND_VEHICLE	Emitter is actually a ground vehicle using a Class B2 transponder with a transmission power < 70W.
10	HAS_1090_ES_IN	Emitter has receive capability for 1090MHz Extended Squitter transmissions.
11	HAS_UAT_IN	Emitter has receive capability for UAT (978MHz Universal Access Transceiver) transmissions.
12	TCAS_OPERATIONAL	Emitter has a functional TCAS (Traffic Collision Avoidance System) onboard.
13	SINGLE_ANTENNA	Emitter is using a single antenna, instead antennas above and below the fuselage. Transmissions may be weak or irregular during maneuvering.
14	DIRECTION_IS_HEADING	Surface position messages provided by the aircraft indicate a heading and not a track angle.
15	HEADING_USES_MAGNETIC_NORTH	Heading reported by the aircraft while on the surface uses magnetic north instead of true north.
16	IDENT	The aircraft has its SPI (Special Position Identification) bits set in Mode A/C or Mode S messages. This indicates that the pilot has depressed the momentary IDENT switch on their transponder, most likely at the request of air traffic control.
17	ALERT	The aircraft is issuing either a permanent or momentary alert. This could correspond to an operational mode change or something else.
18	TCAS_RA	The aircraft has an active TCAS resolution advisory (i.e. the aircraft is warning the pilot to take action in order to avoid colliding with another aircraft).
19	IS_NON_TRANSPONDER	This aircraft is being rebroadcast (not received from a transponder).
20-22 Reserved		
23	UPDATED_BARO_ALTITUDE	Barometric altitude has been updated within the last reporting interval.
24	UPDATED_GNSS_ALTITUDE	GNSS altitude has been updated within the last reporting interval.
25	UPDATED_POSITION	Position (latitude / longitude) has been updated within the last reporting interval.
26	UPDATED_DIRECTION	Direction has been updated within the last reporting interval.
27	UPDATED_HORIZONTAL_SPEED	Horizontal velocity has been updated within the last reporting interval.
28	UPDATED_BARO_VERTICAL_RATE	Barometric vertical velocity has been updated within the last reporting interval.
29	UPDATED_GNSS_VERTICAL_RATE	GNSS vertical velocity has been updated within the last reporting interval.
30-31 Reserved		

5.1.2.2 ECAT Field

The ECAT field indicates the Emitter Category (i.e. airframe type) for each ADSB emitter that is being tracked. This field contains information about what kind of aircraft, ground vehicle, obstacle, or other airspace user is emitting ADS-B packets, and can be used to understand the emitter's maneuvering capability and potential for wake vortex impact.

ECAT Value	Emitter Category
-1	Invalid
0	No Category Information
1	Light Aircraft (< 7,000kg)
2	Medium 1 (7,000kg – 34,000kg)
3	Medium 2 (34,000kg – 136,000kg)
4	High Vortex Aircraft
5	Heavy (> 136,000kg)
6	High Performance (> 5 G acceleration and > 400 kts speed)
7	Rotorcraft
8	Reserved
9	Glider / Sailplane
10	Lighter Than Air
11	Parachutist / Skydiver
12	Ultralight / Hang Glider / Paraglider
13	Reserved 1
14	Unmanned Aerial Vehicle
15	Space / Transatmospheric Vehicle
16	Reserved 2
17	Surface Emergency Vehicle
18	Surface Service Vehicle
19	Point Obstacle
20	Cluster Obstacle
21	Line Obstacle

5.1.2.3

5.1.2.4 NICNAC Bitfield

<table>
<tr><th colspan="16">NICNAC Bitfield</th></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="2">Source Integrity Level (SIL)</td><td colspan="2">Geometric Vertical Accuracy (GVA)</td><td colspan="4">Navigation Accuracy Category: Position (NAC_p)</td><td colspan="3">Navigation Accuracy Category: Velocity (NAC_v)</td><td>Navigation Integrity Category: Barometer (NIC_{baro})</td><td colspan="4">Navigation Integrity Category (NIC)</td></tr>
</table>

NICNAC[0-3]: Navigation Integrity Category (NIC)

The radius of containment (NIC) indicates how much trust should be placed in an aircraft's reported location in the horizonal plane. The NIC reports a radius of containment specified by the avionics system of the emitter. The probability that the aircraft is outside of this radius of containment due to its avionics system receiving a faulty signal from one of its inputs (without displaying an error) is provided by another bitfield called the Source Integrity Level (SIL). Combined, the NIC and SIL indicate how likely it is that an aircraft is not actually contained by a bubble of a specified size, centered at the aircraft's reported location, assuming that the avionics onboard the aircraft are functioning correctly but may be given faulty inputs.

A higher NIC value indicates more trust in an aircraft's reported latitude / longitude position.

NIC Value	0	1	2	3	4	5	6	7	8	9	10	11
Radius of Containment	Unknown	< 20 NM	< 8 NM	< 4 NM	< 2 NM	< 1 NM	< 0.6 NM	< 0.2 NM	< 0.1 NM	< 75 m	< 25 m	< 7.5 m

NICNAC [4]: Navigation Integrity Category: Barometer (NIC_{baro})

The barometric altitude integrity (NIC_{baro}) indicates how much trust should be placed in an aircraft's reported altitude. The field is a single bit that indicates whether the aircraft uses an altimeter that has been cross-checked against other sources. Old school encoding altimeters have many parallel wires and output an altitude in a format called a Gillham Code, and have no built-in method of error checking. A single faulty wire can result in erroneous readings, so this bit lets air traffic control know whether to take altitude readings from the aircraft with a grain of salt. A 0 indicates that the transponder is outputting altitude from a Gillham coded source (with no way to cross check the value), while a 1 indicates that the transponder is outputting altitude from a Gillham coded source while using another sensor to cross-check it, or is using a more modern barometer that supports a protocol with built-in error checking.

A higher NIC_{baro} value (i.e. 1 instead of 0) indicates more trust in the aircraft's reported altitude.

NIC_{baro} Value	0	1
Barometric Altitude Integrity	Altitude is from a Gillham-coded input, not cross checked.	Altitude is from a Gillham-coded input that is being cross-checked with another source, or from a non Gillham-coded input with built-in error checking features.

NICNAC [5-7]: Navigation Accuracy Category: Velocity (NAC_v)

The horizontal velocity error (NAC_v) indicates the expected accuracy of the reported velocity of the aircraft when systems are operating nominally. This varies depending on the accuracy capabilities of the measurement equipment onboard the aircraft, and not how often we expect said equipment to fail.

A higher NAC_v indicates a more accurate velocity measurement system.

NAC_v Value	**Horizontal Velocity Error**
0b000	Unknown or ≥ 10 m/s
0b110	< 10 m/s
0b010	< 3 m/s
0b011	< 1 m/s
0b100	< 0.3 m/s

NICNAC [8-11]: Navigation Accuracy Category: Position (NAC_p)

The estimated position uncertainty (NAC_p) indicates the expected accuracy of the aircraft's reported location when systems are operating nominally. This varies depending on the capabilities of the aircraft's positioning system and not on how often we expect said equipment to fail.

A higher NAC_p indicates a more accurate positioning system.

NAC_p Value	0	1	2	3	4	5	6	7	8	9	10	11
Estimated Position Uncertainty	Unknown or ≥ 10NM	< 10 NM	< 4 NM	< 2 NM	< 1 NM	< 0.5 NM	< 0.3 NM	< 0.1 NM	< 0.5 NM	< 30 m	< 10 m	< 3 m

NICNAC[12-13]: Geometric Vertical Accuracy (GVA)

The geometric vertical accuracy indicates the 95% confidence interval (vertical figure of merit) provided by the aircraft's onboard GNSS system (i.e. assuming some distribution of altitudes, the aircraft's GNSS system is confident that 95 out of 100 times, the aircraft falls within some height of the reported geometric altitude.

GVA Value	0	1	2	3
95% Vertical Figure of Merit (VFOM)	Unknown or ≥ 150 m	< 150 m	≤ 45 m	< 45 m (Was previously "reserved", the actual value of this field may change but is guaranteed to be < 45 m).

NICNAC[14-15]: Source Integrity Level (SIL)

The source integrity level indicates the probability that the aircraft exceeds the bounds of its horizontal radius of containment (NIC) due to a silent fault in signals received by the aircraft (no avionics failure).

SIL Value	Probability of Exceeding NIC Radius of Containment Due to Silent Fault
0	Unknown or > $1x10^{-3}$ per flight hour.
1	≤ $1x10^{-3}$ per flight hour.
2	≤ $1x10^{-5}$ per flight hour.
3	≤ $1x10^{-7}$ per flight hour.
4	Unknown or > $1x10^{-3}$ per sample.
5	≤ $1x10^{-3}$ per sample.
6	≤ $1x10^{-5}$ per sample.
7	≤ $1x10^{-7}$ per sample.

5.1.2.5 ACDIMS Bitfield

ACDIMS Bitfield															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GNSS Offset Forward of Reference Point (GOF)								GNSS Offset Right of Reference Point (GOR)							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Aircraft Length (ACL)							Aircraft Width (ACW)							System Design Assurance (SDA)	

ACDIMS[0-1]: System Design Assurance (SDA)

The system design assurance indicates how robust the aircraft's position reporting systems are to failures of various severities. For instance, SDA = 1, a low SDA value, corresponds to Software and Hardware Design Assurance Level D, which states that a minor failure could cause the aircraft to transmit misleading position information with a probability of $\leq 1x10^{-3}$ per flight hour. A more robust system with SDA = 3, corresponding to Software and Hardware Design Assurance Level B, is expected to transmit misleading position information with a probability of $1x10^{-7}$ per flight hour even under a Hazardous failure condition.

Software Design Assurance categories used in this field are classified under RTCA DO-178B, Airborne Electronic Hardware Design Assurance are classified under RTCA DO-254, and failure classification levels are defined in FAA Advisory Circular AC-23.1309-1E.

SDA Value	Supported Failure Condition	Probability of Undetected Fault causing transmission of False or Misleading Information	Software and Hardware Design Assurance Level
0	Unknown / No Safety Effect	$> 1x10^{-3}$ per flight hour or unknown.	N/A
1	Minor	$\leq 1x10^{-3}$ per flight hour.	D
2	Major	$\leq 1x10^{-5}$ per flight hour.	C
3	Hazardous	$\leq 1x10^{-7}$ per flight hour.	B

ACDIMS[2-8] Aircraft Width (ACW)

Approximate value for the width of the aircraft bounding box, in meters. Can be interpreted as a 7-bit unsigned integer.

ACDIMS[9-15] Aircraft Length (ACL)

Approximate value for the length of the aircraft bounding box, in meters. Can be interpreted as a 7-bit unsigned integer.

ACDIMS[16-23] GNSS Offset Right of Reference Point (GOR)

Signed 8-bit integer representing the left/right offset of the GNSS antenna from the aircraft's reference point. A positive value indicates that the GNSS antenna is to the right of the aircraft reference point.

ACDIMS[24-31] GNSS Offset Forward of Reference Point (GOF)

Signed 8-bit integer representing the forward/back offset of the GNSS antenna from the aircraft's reference point. A positive value indicates that the GNSS antenna is forward of the aircraft's reference point. Always 0 for Mode S aircraft.

5.1.2.6 CRC Field

CSBee messages use a 16-bit Cyclical Redundancy Checksum (CRC-16), which can be calculated using the algorithm in the C++ code snippet below. Note the “swap16” helper function which also needs to be included.

```
uint16_t swap16(uint16_t value) { return (value << 8) | (value >> 8); }

uint16_t CalculateCRC16(const uint8_t *data_p, int32_t length) {
    uint8_t x;
    uint16_t crc = 0xFFFF;
    while (length--) {
        x = crc >> 8 ^ *data_p++;
        x ^= x >> 4;
        crc = (crc << 8) ^ ((uint16_t)(x << 12)) ^ ((uint16_t)(x << 5)) ^ ((uint16_t)x);
    }
    return swap16(crc);
}
```

5.1.3 UAT Aircraft Message

```
#U:ICAO,UAT_FLAGS,CALL,SQUAWK,ECAT,LAT,LON,BARO_ALT,GNSS_ALT,DIR,SPEED,BARO_VRATE,GNSS_VR
ATE,UAT_EMERG,NICNAC,ACDIMS,VERSION,SIGS,SIGQ,UATFPS,CRC\r\n
```

Field	Description	Format	Example value
#U	UAT aircraft message start indicator.		
ICAO	First Digit (optional): Address Qualifier (1 digit: see table 5.1.3.1) Last 6 Digits: ICAO address of aircraft (6 hex digits).	Hex Integer	13C65AC 3C65AC
UAT_FLAGS	Flags bitfield, see table 5.1.3.2.	Hex Integer	12F356A8
CALL	Callsign of aircraft.	String	N61ZP
SQUAWK	SQUAWK of aircraft.	Octal Integer	7232
ECAT	Emitter category, see table 5.1.2.2.	Integer	14
LAT	Latitude, in degrees.	Float	57.57634
LON	Longitude, in degrees.	Float	17.59554
BARO_ALT	Barometric altitude, in feet.	Integer	5000
GNSS_ALT	Geometric altitude, in feet.	Integer	5000
DIR	Direction of aircraft, in degrees [0,360). Consult bit flags to determine whether this field is true heading, magnetic heading, or track.	Integer	35
SPEED	Horizontal velocity of aircraft, in knots.	Integer	464
BARO_VRATE	Barometric vertical velocity of aircraft, in ft/min.	Integer	-1344
GNSS_VRATE	Geometric vertical velocity of the aircraft, in ft/min	Integer	-1344
UAT_EMERG	Aircraft UAT emergency / priority status. See table.	Integer	0
NICNAC	Aircraft data integrity. See table 5.1.2.4	Hex Integer	
ACDIMS	Aircraft physical dimensions and system design assurance. See table 5.1.2.5.	Hex Integer	31BE89F2
VERSION	UAT protocol version used by the aircraft.	Integer	1
SIGS	Signal strength, in dBm.	Integer	-92
SIGQ	Number of bits corrected using UAT forward error correction. 0 (best), 6 (worst).	Integer	2
UATFPS	Number of valid UAT frames received from the aircraft in the last second.	Integer	1
CRC	CRC16 (described in 5.1.2.6).	Hex Integer	2D3E

NOTE: See Mode S Aircraft message decoding reference for any bitfields not prefixed with "UAT_".

5.1.3.1 UAT Address Qualifier

Address Qualifier Value	Meaning
0	Contact is ADSB target (broadcast by transponder) with 24-bit ICAO address.
1	Contact is ADSB target (broadcast by transponder) with self-assigned temporary address.
2	Contact is TIS-B target (broadcast by ground station), using a 24-bit ICAO address.
3	Contact is TIS-B target (broadcast by ground stations) with track file identifier.
4	Contact is a surface vehicle.
5	Contact is a fixed ADSB beacon.

5.1.3.2 UAT_FLAGS Bitfield

Bit	Bit Name	Meaning if the bit is set (1)
0	IS_AIRBORNE	Emitter is airborne.
1	BARO_ALTITUDE_VALID	Emitter has provided a barometer altitude.
2	GNSS_ALTITUDE_VALID	Emitter has provided a GNSS altitude.
3	POSITION_VALID	Emitter has provided a pair of even and odd Compact Position Reporting packets that were decoded to a location.
4	DIRECTION_VALID	Emitter has provided a direction (may be ground track, magnetic compass heading, or true compass heading).
5	HORIZONTAL_SPEED_VALID	Emitter has provided a horizontal velocity.
6	BARO_VERTICAL_RATE_VALID	Emitter has provided a barometric vertical velocity.
7	GNSS_VERTICAL_RATE_VALID	Emitter has provided a GNSS vertical velocity.
8	HAS_CDTI	Aircraft has Cockpit Display of Traffic Information (CDTI) equipped.
9	TCAS_OPERATIONAL	Aircraft is equipped with an operational TCAS / ACAS system.
10	DIRECTION_IS_HEADING	Surface position messages provided by the aircraft indicate a heading and not a track angle.
11	HEADING_USES_MAGNETIC_NORTH	Heading reported by the aircraft while on the surface uses magnetic north instead of true north.
12	IDENT	The aircraft has its SPI (Special Position Identification) bits set in Mode A/C or Mode S messages. This indicates that the pilot has depressed the momentary IDENT switch on their transponder, most likely at the request of air traffic control.
13	IS_UTC_COUPLED	The aircraft is transmitting frames precisely aligned to the UTC second.
14 Reserved		
15	TCAS_RA	The aircraft has an active TCAS resolution advisory (i.e. the aircraft is warning the pilot to take action in order to avoid colliding with another aircraft).
16	RECEIVING_ATC_SERVICES	
17-22 Reserved		
23	UPDATED_BARO_ALTITUDE	Barometric altitude has been updated within the last reporting interval.
24	UPDATED_GNSS_ALTITUDE	GNSS altitude has been updated within the last reporting interval.
25	UPDATED_POSITION	Position (latitude / longitude) has been updated within the last reporting interval.
26	UPDATED_DIRECTION	Direction has been updated within the last reporting interval.
27	UPDATED_HORIZONTAL_SPEED	Horizontal velocity has been updated within the last reporting interval.
28	UPDATED_BARO_VERTICAL_RATE	Barometric vertical velocity has been updated within the last reporting interval.
29	UPDATED_GNSS_VERTICAL_RATE	GNSS vertical velocity has been updated within the last reporting interval.
30-31 Reserved		

5.1.3.3 UAT_EMERG Bitfield

Contains the emergency / priority status reported by a UAT emitter.

UAT_EMERG Value	Meaning
0	No emergency.
1	General emergency.
2	Lifeguard / medical emergency.
3	Minimum fuel.
4	No communications.
5	Unlawful interference.
6	Downed aircraft.
7	Reserved.

5.1.4 Statistics Message

This message contains some useful statistics about ADSBee's operational status. Format of that frame is shown below:

```
#S:VERSION,DPS,RAW_SFPS,SFPS,RAW_ESFPS,ESFPS,RAW_UATFPS,UATFPS,TSCAL,UPTIME,CRC\r\n
```

Field	Description	Format	Example Value
#S	Statistics message start indicator.		
VERSION	Version of the CSBee protocol being used.	Integer	3
SDPS	Number of attempted Mode S demodulations in the last second.	Integer	106
RAW_SFPS	Number of squitter (56-bit Mode S) frames received in the last second.	Integer	150
SFPS	Number of valid squitter (56-bit Mode S) frames received in the last second.	Integer	20
RAW_ESFPS	Number of extended squitter (112-bit Mode S) frames received in the last second.	Integer	15
ESFPS	Number of valid Mode S frames received in the last second.	Integer	3
RAW_UATFPS	Number of UAT frames received in the last second.	Integer	5
UATFPS	Number of valid UAT frames received in the last second.	Integer	4
NUM_AIRCRAFT	Number of aircraft tracked in the last second.	Integer	20
TSCAL	Calibration value for TS field in raw frames	Integer	13999415
UPTIME	Receiver uptime, in seconds.	Integer	13456
CRC	CRC16 (described in 5.1.4.1).	Hex Integer	2D3E

5.1.4.1 CRC Field

See 5.1.2.6.

5.2 GDL90

ADSBee complies with the GDL90 public data interface specification 560-1058-00 Rev A. The GDL90_NO_UAT_UPLINK protocol variant is identical to GDL90, except that UAT uplink messages are not forwarded.

5.3 MAVLINK

Tracked aircraft information is sent in MAVLINK ADSB_VEHICLE messages, in a data burst once per second. The data burst consists of 1x MAVLINK Heartbeat message, Nx ADSB_VEHICLE messages, where N is the number of tracked aircraft, and 1x special delimiter message (MAVLINK_REQUEST_DATA_STREAM for MAVLINK1 mode, MAVLINK_MESSAGE_INTERVAL for MAVLINK2 mode) which indicates the end of the list of tracked aircraft. Note that this is a binary protocol which is not human readable.

WARNING: Windows has a bug, which causes some machines to recognize a serial port reporting MAVLINK packets as a mouse, which can result in phantom mouse movements and clicks (ask me how I know). Please use caution while running MAVLINK on a serial port while the computer is unattended.

5.3.1 MAVLINK_HEARTBEAT (Message ID 0) Packet Definition

Field Name	Type	Units	Values	Description
custom_mode	`uint32_t`		0	Unused.
type	`uint8_t`		MAV_TYPE_ADSB	Always MAV_TYPE_ADSB (27).
autopilot	`uint8_t`		MAV_AUTOPILOT_INVALID	Always MAV_AUTOPILOT_INVALID (8).
base_mode	`uint8_t`		0	Unused.
system_status	`uint8_t`		MAV_STATE_ACTIVE	Always MAV_STATE_ACTIVE (4).
mavlink_version	`uint8_t`		MAVLINK_VERSION	1 for MAVLINK1 mode, 2 for MAVLINK2 mode.

5.3.2 MAVLINK ADSB_VEHICLE (Message ID 246) Packet Definition

From: https://mavlink.io/en/messages/common.html#ADSB_VEHICLE

Field Name	Type	Units	Values	Description
ICAO_address	`uint32_t`			ICAO address
lat	`int32_t`	degE7		Latitude
lon	`int32_t`	degE7		Longitude
altitude_type	`uint8_t`		ADSB_ALTITUDE_TYPE	ADSB altitude type.
altitude	`int32_t`	mm		Altitude(ASL)
heading	`uint16_t`	cdeg		Course over ground
hor_velocity	`uint16_t`	cm/s		The horizontal velocity
ver_velocity	`int16_t`	cm/s		The vertical velocity. Positive is up
callsign	`char[9]`			The callsign, 8+null
emitter_type	`uint8_t`		ADSB_EMITTER_TYPE	ADSB emitter type.
tslc	`uint8_t`	s		Time since last communication in seconds
flags	`uint16_t`		ADSB_FLAGS	Bitmap to indicate various statuses including valid data fields
squawk	`uint16_t`			Squawk code

5.3.3 MAVLINK_REQUEST_DATA_STREAM (Message ID 66) Packet Definition

This message serves as the delimiter which indicates the end of the aircraft list in MAVLINK1 mode.

Field Name	Type	Units	Description
req_message_rate	`uint16_t`		Always 0 (unused).
target_system	`uint8_t`		Always 0 (unused).
target_component	`uint8_t`		Always 0 (unused).
req_stream_id	`uint8_t`		Always 0 (unused).
start_stop	`uint8_t`		Always 0 (unused).

5.3.4 MAVLINK MESSAGE_INTERVAL (Message ID 244) Packet Definition

This message serves as the delimiter which indicates the end of the aircraft list in MAVLINK2 mode.

From: https://mavlink.io/en/messages/common.html#MESSAGE_INTERVAL

Field Name	Type	Units	Description
message_id	`uint16_t`		The ID of the requested MAVLink message. v1.0 is limited to 254 messages. NOTE: For ADSBee m1421, message_id is always 246, corresponding to the ADSB_VEHICLE message.
interval_us	`int32_t`	us	The interval between two messages. A value of -1 indicates this stream is disabled, 0 indicates it is not available, > 0 indicates the interval at which it is sent. NOTE: For ADSBee m1421, message_id is always 1000 us.

5.4 Mode S Beast

Mode S Beast is a popular feed protocol used by both open-source and commercial data exchanges to send Mode S data, often over TCP sockets or serial connections.

Mode S Beast sends packets as raw binary, with 0x1a used as an “escape” character indicating the start of a Beast frame. Any non start-of-frame instance of 0x1a is escaped with an additional 0x1a beforehand. Thus, 0x1a can be interpreted as the start of a Beast frame, and 0x1a1a can be interpreted as a 0x1a value within a Beast frame.

5.4.1 Beast: Mode S Frames

Field	Frame Start	Frame Type	Timestamp	RSSI	Payload
Length [Bytes]	1	1	6-12	1-2	Mode S Short: 7-14 Mode S Long: 14-28
Example	0x1a	0x32	0x123456781a1aff	0x56	<data>
Definition	0x1a Frame start is always an escape char.	0x32: Mode S Short 0x33: Mode S Long No escapes.	6-byte 12MHz MLAT timestamp, MSB first. May have up to 6 escapes.	RSSI byte is “Beast Power Level”, which is the square root of the un-logged dBFS value. May have up to 1 escape.	Packet contents. May theoretically have up to <packet_length> escapes, but this would not correspond to a valid packet and will not be seen in practice.

5.4.2 Beast: UAT Frames

Field	Frame Start	Frame Type	UAT Message Type	Timestamp	RSSI	Payload
Length [Bytes]	1	1	1	6-12	1-2	UAT Short ADSB:
Example	0x1a	0xec	0x73	0x123456781a1aff	0x56	<data>
Definition	0x1a Frame start is always an escape char.	0xec Indicates frame type UAT (both ADSB and uplink). No escapes.	0x73 (‘s’): UAT short ADSB (30 byte payload, not including escapes). 0x5c (‘l’): UAT long ADSB (48 byte payload, not including escapes) 0x75 (‘u’): UAT Uplink (552 byte payload, not including escapes) No escapes.	6-byte 12MHz MLAT timestamp, MSB first. May have up to 6 escapes.	RSSI byte is “Beast Power Level”, which is the square root of the un-logged dBFS value. May have up to 1 escape.	Packet contents, including both payload and FEC parity. Uplink packets remain interleaved (transmitted byte order retained). May theoretically have up to <packet_length> escapes, but this would not correspond to a valid packet and will not be seen in practice.

5.5 Raw Packets

5.5.1 Raw Mode S Frames

Raw Mode S packets are reported in the format below. Only packets that have passed checksum validation and match a supported Mode S downlink format are reported.

```
#MDS*RAW_FRAME;(SOURCE,SIGS,SIGQ,TS)\r\n
```

Field	Description	Format	Example value
#MDS	Indicates the start of a raw Mode S frame.		
RAW_FRAME	The packet contents as raw hexadecimal. Varies in length depending on packet length.	Hexadecimal String	8D48C22D60AB0452BFAD19A695E0
SOURCE	An unsigned integer	`int16_t`	2
SIGS	Signal strength of the packet in dBm, printed as a decimal value.	`int16_t`	-60
SIGQ	Signal quality of the packet in dB, printed as a decimal value.	`int16_t`	2
TS	64-bit 48MHz MLAT counter ticks, printed as a hexadecimal string.	`uint64_t`	00000000FB671342

5.5.2 Raw UAT ADSB Frames

Raw UAT frames are reported in the format below. Only packets that have passed checksum validation are reported. Note that UAT uplink frames can be quite long (552 Bytes / 1104 chars), so serial buffers need to be able to handle this!

```
#UAT*-RAW_FRAME;(SIGS,SIGQ,TS)\r\n
```

Field	Description	Format	Example value
#UAT	Indicates the start of a raw UAT frame.		
RAW_FRAME	The packet contents as raw hexadecimal. Varies in length depending on packet length.	Hexadecimal String	
SIGS	Signal strength of the packet in dBm.	`int16_t`	-60
SIGQ	Signal quality of the packet in dB.	`int16_t`	2
TS	64-bit MLAT counter value in hexadecimal format.	`uint64_t`	00000000FB671342

5.5.3 Raw UAT Uplink Frames

Raw UAT uplink frames are reported in the format below. Only packets that have passed checksum validation are reported. Note that UAT uplink frames can be quite long (552 Bytes / 1104 chars), so serial buffers need to be able to handle this!

```
#UAT*+RAW_FRAME;(SIGS,SIGQ,TS)\r\n
```

Field	Description	Format	Example value
#UAT	Indicates the start of a raw UAT frame.		
RAW_FRAME	The packet contents as raw hexadecimal. Varies in length depending on packet length.	Hexadecimal String	
SIGS	Signal strength of the packet in dBm.	`int16_t`	-60
SIGQ	Signal quality of the packet in dB.	`int16_t`	2
TS	64-bit MLAT counter value in hexadecimal format.	`uint64_t`	00000000FB671342

5.6 Aircraft JSON

The ADSBee m1421 streams real-time aircraft data over the Console UART when its reporting protocol is set to AIRCRAFT_JSON (AT+PROTOCOL_OUT=CONSOLE,AIRCRAFT_JSON). The full aircraft list is sent once per second as one JSON object per tracked aircraft, with objects separated by newline characters.

The ADSBee Aircraft JSON format is intended to be compatible with the JSON format emitted by readsb (https://github.com/wiedehopf/readsb/blob/dev/README-json.md).

5.6.1 Message Format

The Console UART stream contains one or more newline-terminated JSON objects, one per tracked aircraft. Fields are omitted when their data has not been received or is not applicable to the aircraft type; clients must treat absent fields as unknown rather than zero.


```
{"hex":"a1b2c3","type":"adsb_icao","flight":"UAL123 ","lat":37.62345,"lon":-122.37812,"alt_baro":37000,"gs":480,"track":270.0,"baro_rate":0,"nic":8,"nic_baro":1,"nac_p":9,"nac_v":2,"sil":3,"gva":2,"sda":2,"version":2,"rssi":-68,"messages":142}
{"hex":"c3d4e5","type":"adsb_icao","on_ground":1,"lat":37.61200,"lon":-122.38500,"alt_baro":0,"gs":12,"track":90.0,"nic":8,"nic_baro":1,"nac_p":9,"nac_v":2,"sil":3,"gva":2,"sda":2,"version":2,"rssi":-71,"messages":23}
```

5.6.2 JSON Fields

Field	Type	Condition	Mode S	UAT	Units	Description
hex	string	Always	✓	✓	—	6-digit ICAO 24-bit address in lowercase hex. UAT non-ICAO addresses (self-assigned or TIS-B track file) are prefixed with ~.
type	string	Always	✓	✓	—	Aircraft type. Mode S: adsb_icao, adsr_icao (non-transponder), adsb_icao_nt (Class B ground vehicle). UAT: adsb_icao, adsb_other, tisb_icao, tisb_trackfile.
flight	string	If callsign received	✓	✓	—	8-character flight identifier / callsign, right-padded with spaces.
squawk	string	If squawk received	✓	✓	—	4-digit octal transponder squawk code.
category	string	If category received	✓	✓	—	Emitter category in readsb A0-D7 format. Letter A-D = set 0-3; digit 0-7 = subcategory.
lat	number	If position valid	✓	✓	degrees	WGS-84 latitude, 5 decimal places (~1.1 m resolution).
lon	number	If position valid	✓	✓	degrees	WGS-84 longitude, 5 decimal places (~1.1 m resolution).
alt_baro	integer	If baro altitude valid	✓	✓	ft	Barometric pressure altitude.
alt_geom	integer	If GNSS altitude valid	✓	✓	ft	Geometric (GNSS/GPS) altitude.
gs	integer	If speed valid	✓	✓	kt	Ground speed.
track	number	If direction valid, not heading	✓	✓	degrees	True track over ground, 1 decimal place.
mag_heading	number	If heading valid, magnetic	✓	✓	degrees	Magnetic heading, 1 decimal place.
true_heading	number	If heading valid, true north	✓	✓	degrees	True heading, 1 decimal place.
baro_rate	integer	If baro vertical rate valid	✓	✓	ft/min	Barometric vertical rate of climb/descent.

geom_rate	integer	If GNSS vertical rate valid	✓	✓	ft/min	Geometric (GNSS) vertical rate of climb/descent.
nic	integer	Always	✓	✓	—	Navigation Integrity Category (0-11).
nic_baro	integer	Always	✓	✓	—	Navigation Integrity Category Barometric (0-4).
nac_p	integer	Always	✓	✓	—	Navigation Accuracy Category - Position (0-11).
nac_v	integer	Always	✓	✓	—	Navigation Accuracy Category - Velocity (0-4).
sil	integer	Always	✓	✓	—	Surveillance Integrity Level (0-3).
gva	integer	Always	✓	✓	—	Geometric Vertical Accuracy (0-2).
sda	integer	Always	✓	✓	—	System Design Assurance (0-3).
version	integer	Always (Mode S); if >= 0 (UAT)	✓	✓	—	ADS-B version number.
rssi	integer	Always	✓	✓	dBm	Last received signal strength.
messages	integer	Always	✓	✓	—	Total valid messages received from this aircraft since tracking began.
alert	integer (1)	If alert flag set	✓	✗	—	Present (value 1) when the Mode S alert flag is active. Mode S only.
spi	integer (1)	If SPI flag set	✓	✗	—	Present (value 1) when the aircraft is squawking ident (Special Position Identification). Mode S only.
emergency	string	If emergency status set	✗	✓	—	Emergency/priority status: none, general, lifeguard, minfuel, nordo, unlawful, downed, reserved. UAT only.
on_ground	integer (1)	If NOT airborne	✓	✓	—	Present (value 1) when aircraft is on the ground. For UAT, also set for surface vehicles.